

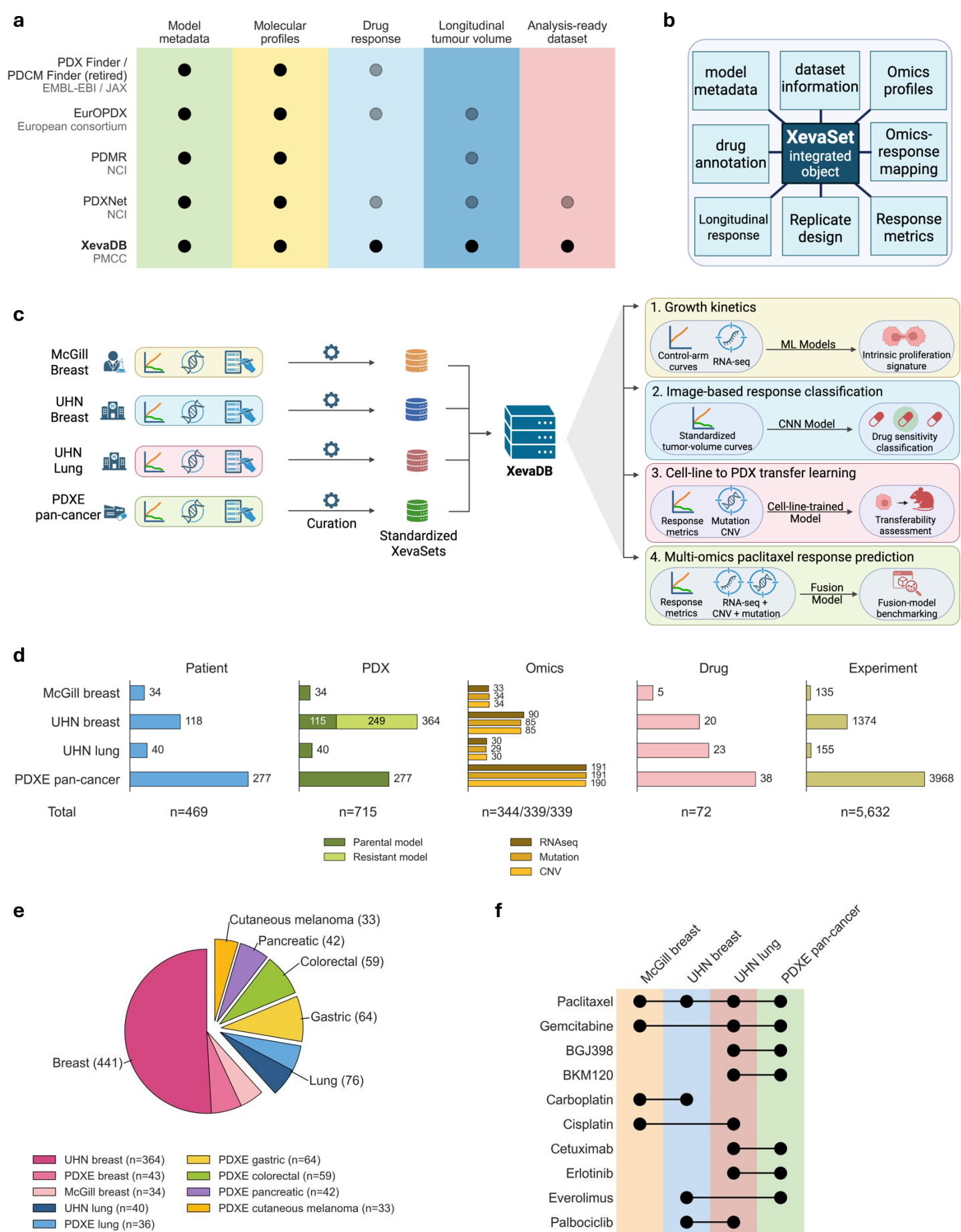


Figure 1. Overview of XevaDB, a standardized longitudinal PDX pharmacogenomic resource.

(a) Comparison of XevaDB with four existing PDX-related resources — PDX Finder/PDCM Finder (retired), EurOPDX, PDMMR, and PDXNet — across five data-availability capabilities: model metadata, molecular profiles, drug response, longitudinal tumour volume, and analysis-ready dataset. Solid dots indicate full availability, semi-transparent dots indicate limited/partial availability, and the absence of a dot indicates the capability is not available. The organization or consortium responsible for each resource is shown below its name (EMBL-EBI, European Molecular Biology Laboratory–European Bioinformatics Institute; JAX, The Jackson Laboratory; NCI, National Cancer Institute; PMCC, Princess Margaret Cancer Centre).

(b) XevaDB's internal data architecture. The XevaSet integrated object unifies model metadata, dataset information, omics profiles, drug annotation, omics-response mapping, longitudinal response, replicate design, and response metrics within a single standardized structure.

(c) Data curation and analysis pipeline. Raw longitudinal response, omics, and clinical metadata from four source cohorts (McGill Breast, UHN Breast, UHN Lung, PDXE pan-cancer) are curated into standardized XevaSets and integrated into XevaDB, which supports four downstream case studies: (1) growth-kinetics modeling from control-arm curves and RNA-seq, (2) image-based drug-response classification from standardized tumour-volume curves, (3) cell-line-to-PDX transfer learning using response metrics and mutation/CNV data, and (4) multi-omics prediction of paclitaxel response.

(d) Summary counts across the four cohorts for patients (n=469), PDX models (n=715; parental and resistant models shown separately), matched omics profiles (RNA-seq/mutation/CNV; n=344/339/339), drugs (n=72), and treatment-control experiments (n=5,632).

(e) Distribution of PDX models by cancer type (n=715 total across six tumour types), with each wedge group further broken down by contributing dataset (shown in the legend); slice groups are radially separated to indicate cancer type.

(f) Drugs shared across more than one dataset (10 of 72 total drugs), shown as a dot-matrix across the four cohorts; connected dots indicate the same drug tested in multiple datasets.

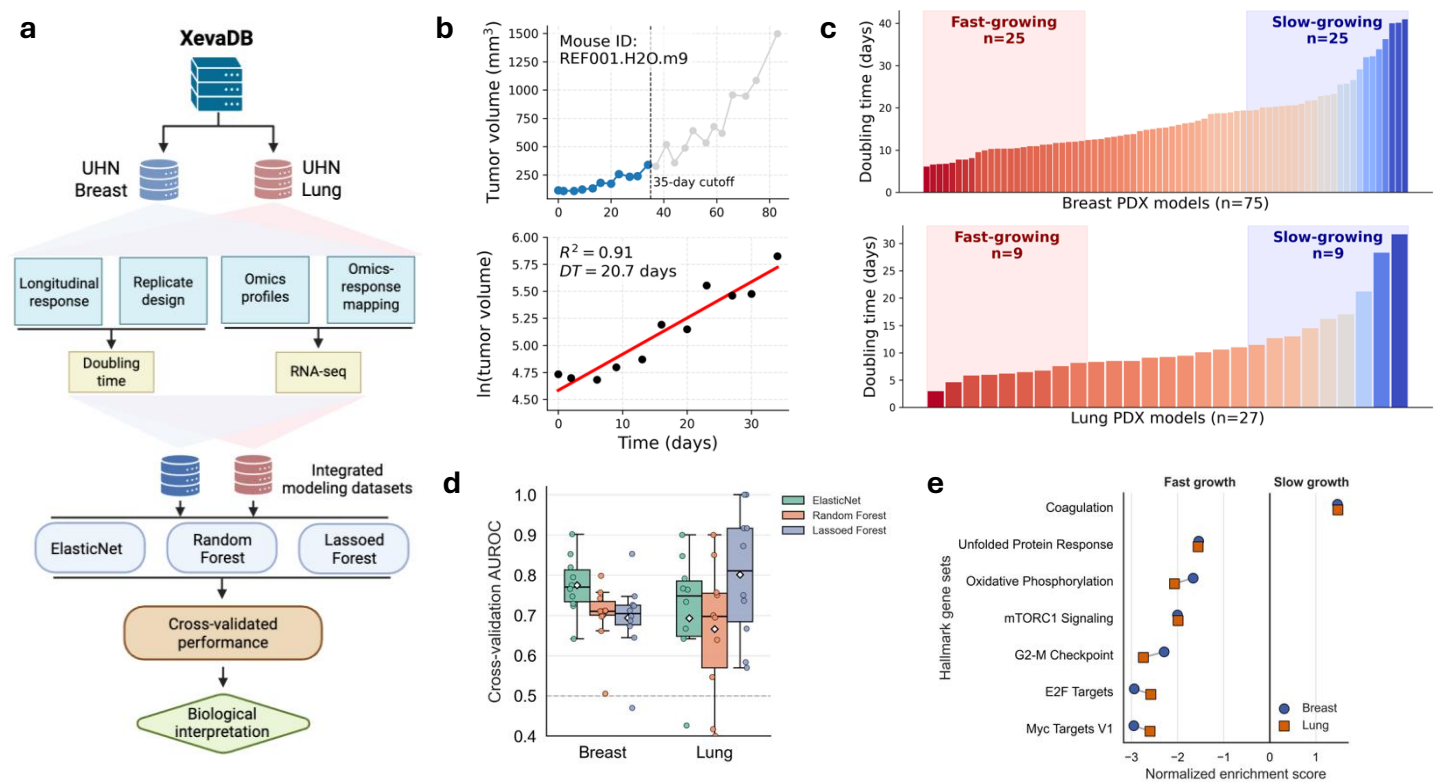


Figure 2. Transcriptomic modeling of intrinsic PDX growth kinetics.

(a) Case study one workflow: control-arm longitudinal tumour-volume curves and matched RNA-seq from breast and lung PDX cohorts are used to estimate model-level doubling time, modeled by ElasticNet, Random Forest, and Lassoed Forest classifiers under repeated cross-validation. (b) Representative doubling-time estimation for one control-arm mouse (REF001.H2O.m9): raw tumour-volume trajectory (top; points within the 35-day fitting window in blue, excluded points in gray) and the corresponding exponential-growth fit ($\ln(\text{tumour volume})$ vs. time, bottom), with goodness-of-fit ($R^2 = 0.91$) and doubling time ($DT = 20.7$ days) annotated. (c) Distribution of estimated doubling times across breast ($n = 75$) and lung ($n = 27$) PDX models (integrated modeling datasets shown in a), with the fastest- and slowest-growing tertiles used to define the classification task highlighted (breast: $n = 25/25$; lung: $n = 9/9$). (d) Cross-validated AUROC of ElasticNet, Random Forest, and Lassoed Forest classifiers distinguishing fast- from slow-growing models, breast and lung; dashed line, chance performance. (e) Cross-cohort comparison of Hallmark gene-set enrichment (GSEA NES) associated with fast- vs. slow-growing tumours in the breast and lung cohorts.

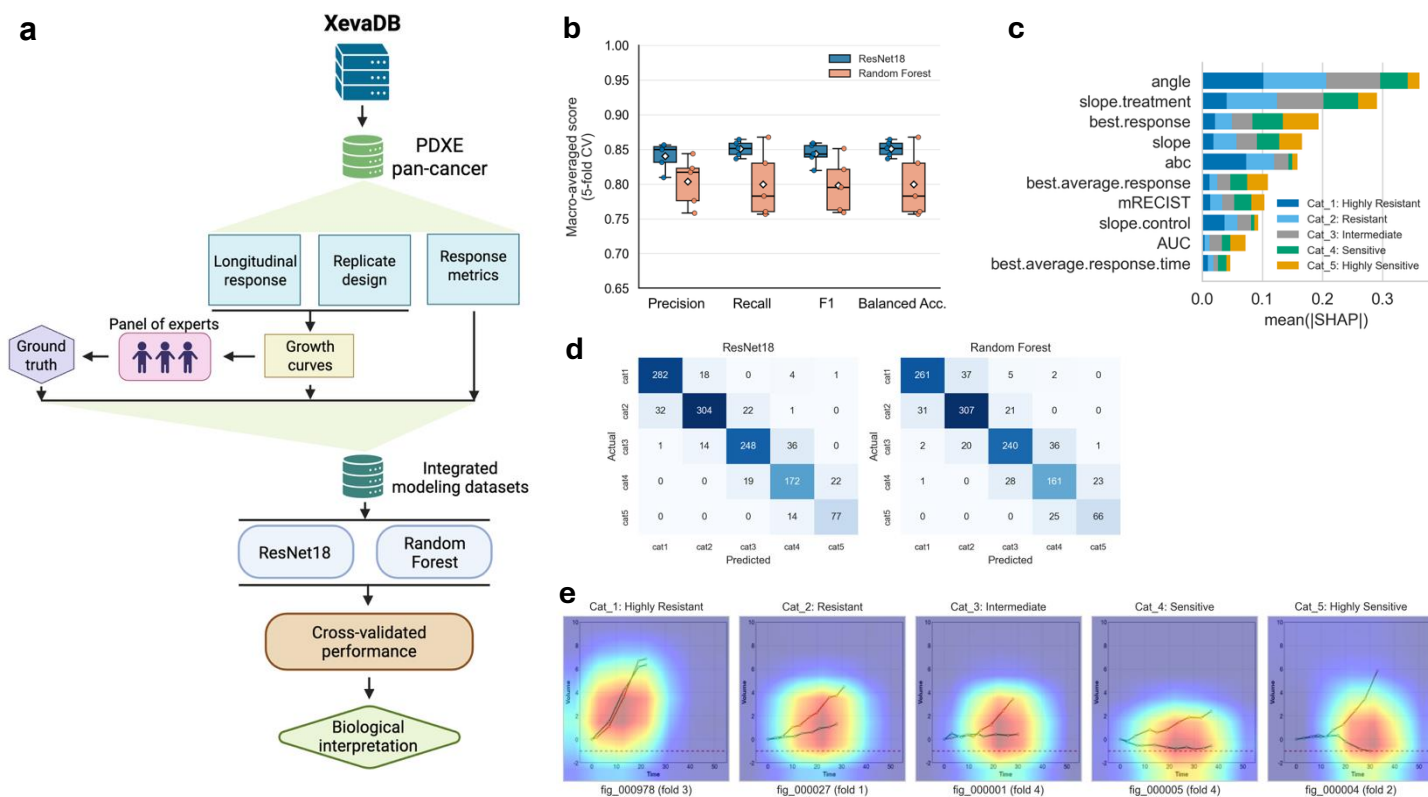


Figure 3. Image-based deep learning captures non-linear drug-response dynamics.

(a) Case study two workflow: tumour-volume growth curves for model-drug batches from the PDxE pan-cancer cohort are rendered as standardized images, scored by a three-rater expert panel into five ordinal response categories, and used to train and compare an image-based ResNet18 classifier against a feature-based Random Forest classifier under matched 5-fold cross-validation. **(b)** Cross-validated performance (macro-averaged precision, recall, F1-score, and balanced accuracy) of ResNet18 versus Random Forest across 5 outer folds. **(c)** Random Forest feature importance (mean absolute SHAP value, stacked by response category) for the top 10 of thirteen engineered response features, ranked by total mean absolute SHAP across categories. **(d)** Pooled out-of-fold confusion matrices for ResNet18 and Random Forest across all 1,267 curves. **(e)** Representative Grad-CAM visualizations for one correctly classified, held-out example per response category.

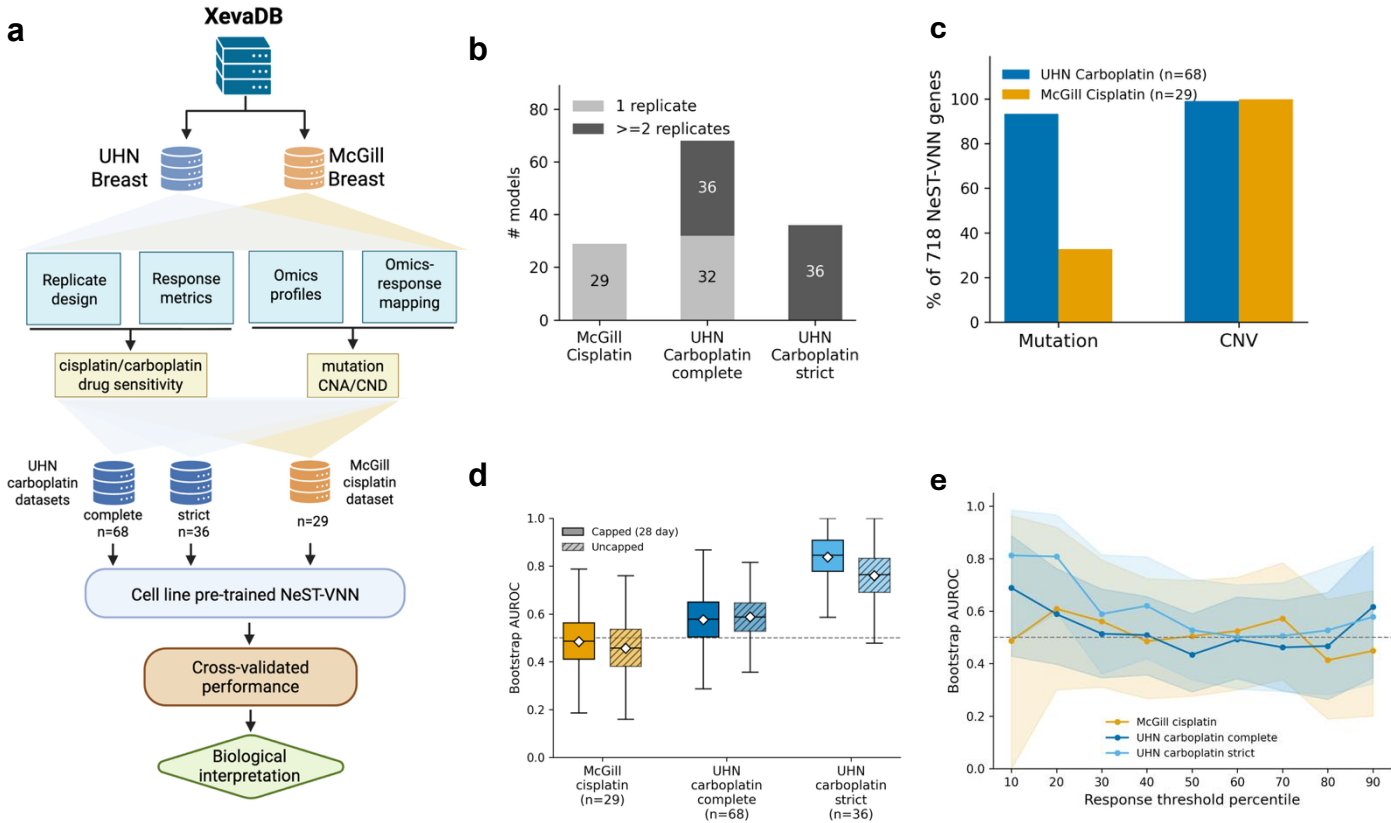


Figure 4. Cross-platform validation of NeST-VNN on independent PDX cohorts.

(a) Case study three workflow: drug-response, mutation, and copy-number data from the McGill (cisplatin, $n = 29$) and UHN (carboplatin; complete cohort $n = 68$, replicate-supported "strict" cohort $n = 36$) breast PDX cohorts are used as external input to a cell-line-pretrained NeST-VNN ensemble, validated against a growth-curve response label. (b) Number of models with exactly one versus two or more treatment-arm replicate mice, for each cohort/cohort definition. (c) Fraction of the 718-gene NeST-VNN input panel with recorded mutation and copy-number (CNV) calls in each cohort's raw molecular profiles. (d) Bootstrap 95% confidence intervals for NeST-VNN's AUROC under the capped (28 day window, primary) and uncapped (full observed duration) response-label definitions, for each of the three cohort definitions; dashed line, chance performance. (e) NeST-VNN performance across alternative response definitions. AUROC was evaluated across cohort-specific treatment-arm slope percentile cutoffs, with models below each cutoff classified as sensitive. Lines represent the McGill cisplatin, UHN carboplatin complete and UHN carboplatin strict cohorts, shaded regions indicate bootstrap 95% confidence intervals, and the dotted line indicates chance performance (AUROC = 0.5).

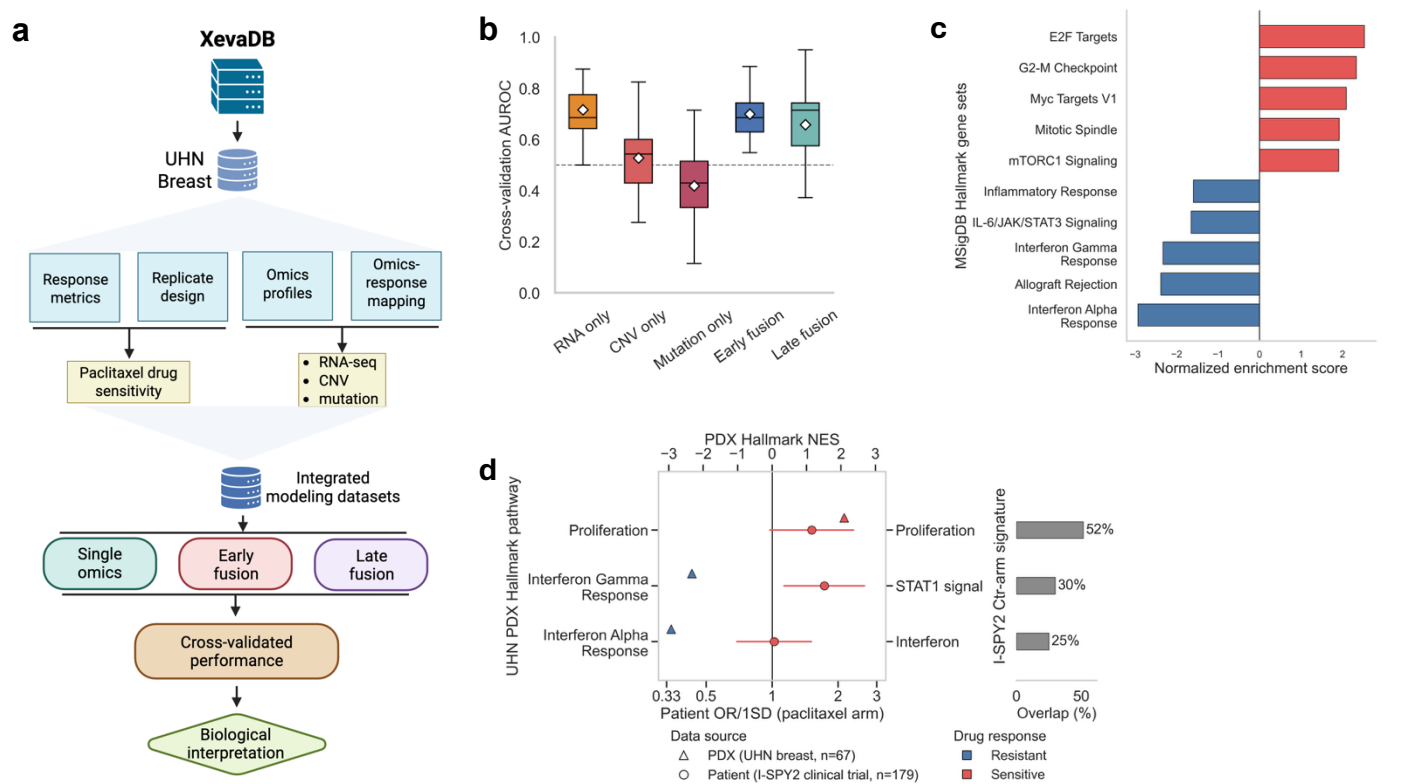


Figure 5. XevaDB-based multi-omics prediction of paclitaxel response in breast cancer PDX models.

(a) Case study four workflow: drug-sensitivity metrics and RNA-seq/CNV/mutation profiles for the UHN breast paclitaxel cohort are extracted from XevaDB and assembled into single-omics, early-fusion, and late-fusion modeling datasets, evaluated by cross-validation. **(b)** Cross-validation AUROC of the random forest classifier across single-omics (RNA-only, CNV-only, mutation-only) and multi-omics (early fusion, late fusion) feature sets; boxes show median/IQR and whiskers across repeated CV folds, diamonds mark the mean, dashed line marks chance (AUROC = 0.5). **(c)** Pre-ranked GSEA of the full RNA transcriptome against MSigDB Hallmark gene sets; top 6 significant pathways per direction shown (FDR $q < 0.05$). Positive NES: enriched in Sensitive models; negative NES: enriched in Resistant models. **(d)** Cross-cohort comparison of transcriptomic programmes associated with paclitaxel response in UHN PDXs ($n=67$) and pathological complete response (pCR) in the I-SPY2 paclitaxel control arm ($n=179$ patients). Triangles, PDX normalized enrichment scores (NES) for MSigDB Hallmark pathways; circles, patient odds ratios (OR) for pCR per one-standard-deviation (1SD) increase in the corresponding predefined I-SPY2 signature, with 95% confidence intervals (refit from patient-level I-SPY2 data; see Methods). Markers are colored by their own direction of association (blue, Resistant; red, Sensitive), so a row is concordant when both markers share a color and discordant when they differ. Positive PDX NES indicates enrichment in sensitive PDXs, whereas patient OR values above 1 indicate higher odds of pCR. "Proliferation" reflects the median NES across five Hallmark gene sets significantly enriched in sensitive PDXs (E2F Targets, G2-M Checkpoint, Mitotic Spindle, Myc Targets V1, Myc Targets V2). Percentages indicate gene overlap between each I-SPY2 signature and its corresponding Hallmark gene set. Not all pathways shown in (c) are evaluated here: mTORC1 Signaling, Allograft Rejection, IL-6/JAK/STAT3 Signaling and Inflammatory Response lacked an I-SPY2 signature with adequate gene-level overlap and are not shown.